Project Proposal: Home Energy Coach

A Personalized Green Habit Recommender for Students and Renters in Shared Flats

Revised version, 16 November 2026

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Frankfurt University of Applied Sciences · High Integrity Systems, Winter Semester 2026/27

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Abstract

This proposal describes the plan for Home Energy Coach, the team's Personalized Green Habit Recommender for the High Integrity Systems Project at Frankfurt University of Applied Sciences (WS 2026/27). The app is aimed at students and renters in shared flats (WGs). They cannot change their building's insulation or heating system, but they can change their daily habits. Users fill in a short energy profile, get a ranked list of habits that suit their flat and schedule, commit to one small goal at a time and are reminded when it is time to act. The main feature is the Green Hours planner. It uses live generation data for the German electricity market from the Bundesnetzagentur's SMARD platform to tell each user the lowest-CO_2 hours for laundry, the dishwasher and charging, limited to the hours they are actually at home. The native iOS app (SwiftUI) talks to a FastAPI back-end, which handles the application logic and passes all data analysis to an R analytics service exposed through the plumber package. That analysis covers the hourly CO_2 forecast, recommendation ranking, association-rule mining, user clustering and the evaluation of nudges. User data is stored in PostgreSQL, and the server side runs in Docker containers. Following the course's green software requirement, the team measures the app's own energy and resource use from the first prototype onwards, reduces it in every sprint and shows the results in the app. The project runs as five Scrum sprints with rotating Scrum Master and Product Owner roles, and all progress is tracked in Jira.

I. Introduction

Students and renters in shared flats (WGs) cannot insulate their building or replace its heating, but they can change their daily habits, including *when* their appliances run. The CO_2 intensity of German electricity changes strongly within a day as wind and solar output rises and falls, and the Bundesnetzagentur publishes the underlying data free of charge on SMARD [1]. Home Energy Coach turns this data and a short user profile into small, personal habits, and uses timely reminders, one-tap check-ins and immediate feedback to make them stick [2, 3].

II. Objectives: What the App Will Do

Table I lists what the finished app will be able to do. Each objective has a target that is measured at the final delivery on 25 January 2027.

Table I
Objectives of Home Energy Coach

#	Objective	What the app does	Measurable target
O1	Personal habit plan	After an onboarding quiz of 12 questions (flat size, flatmates, heating and hot-water type, appliances, weekly at-home hours), the user gets a top-5 list of habits that are possible in their flat. Each habit shows its saving in kWh, kg CO_2 and euro and a one-sentence "why this" reason.	0 infeasible habits recommended for 200 synthetic test profiles; Precision@5 higher than a popularity baseline in 5-fold cross-validation
O2	Green Hours planner	By 06:00 every day, the app shows today's and tomorrow's hourly CO_2 curve of the German grid and the user's three greenest at-home slots for each of laundry, dishwasher and charging, with a reminder 30 minutes before each slot.	Forecast greenest 3-hour window overlaps the actual greenest window on at least 70 % of days in a 4-week back-test
O3	Habit loop	The user commits to one habit at a time ("on Saturday at 12:00 I start the laundry"), confirms it with one tap and sees the CO_2 saved and the streak at once. After 7 on-time check-ins in a row, the reminders stop and the next habit is proposed.	At least 2 of 3 pilot users keep a habit with a check-in rate of at least 70 % over two weeks
O4	Progress overview	A dashboard shows estimated monthly kWh, CO_2 and euro per end use (heating, hot water, appliances), weekly trends, optional meter readings and a checklist of one-off actions such as LED bulbs or a green tariff.	Every figure shown is traceable to a documented source (SMARD, UBA, Stromspiegel)
O5	Shared flat	Flatmates join a WG group with an invite code, agree on shared goals and compete with other flats. Only the flat's total score is shown, never who did what.	A group of 6 members passes the end-to-end test from invitation to challenge result
O6	Privacy and usability	Users see what data is stored, export or delete it, and mute reminders per habit.	Mean System Usability Scale score of at least 68 in the pilot [4]; export and deletion pass end-to-end tests
O7	Green app	The app measures its own energy and data use and shows it on the screen "How green is this app?".	Median server CPU time and data per user session at least 30 % below the Sprint 4 baseline; "Low" energy impact in Xcode

The app has eleven screens beyond login and basic data entry: onboarding quiz, dashboard, recommendation feed, commitment and check-in, Green Hours planner, habit library (22 habits), one-time checklist, WG challenge, trends, privacy centre and "How green is this app?". The course requirements are covered as follows: all analysis runs in R (Section III), eleven interfaces are planned where eight are required, the app's own footprint is objective O7, and Scrum with rotating roles is tracked in Jira (Section V).

Out of scope: Android and web clients, control of smart plugs or thermostats, automatic smart-meter integration, electricity grids outside Germany, and a public App Store release (the pilot uses TestFlight).

III. What R and plumber Are Used For

All data analysis runs in R [5]. The `plumber` package [6] is only the interface: it makes the R functions callable over HTTP by the FastAPI back-end, and the iOS app never calls R directly. R performs the six tasks in Table II. Three of them are machine-learning or data-mining models (P3, P4, P5); the Green Hours forecast (P2) is a calculation and needs no trained model. Until real users exist, the models are trained on synthetic users built from Stromspiegel consumption data [7], UBA end-use shares [8] and REFIT appliance run times [9], together with a student survey in Sprint 2; pilot data replace them in Sprint 5.

TABLE II
TASKS OF THE R SERVICE

#	Task	Method (R packages)	Runs
P1	Load grid data	Download the actual generation per source and the day-ahead forecast from SMARD; convert quarter-hours to hours in Europe/Berlin time and fill short gaps (`httr2`, `dplyr`, `lubridate`)	Nightly
P2	Green Hours forecast	SMARD's day-ahead forecast gives total, wind and solar generation. The remaining "other" generation is multiplied by its emission factor, estimated from the actual mix of the last 7 days and the UBA factors per source [10]. The resulting g CO_2/kWh per hour is matched to the user's at-home hours.	Nightly; after a schedule change
P3	Habit ranking	Filter by the flat's equipment and score by CO_2 saving, ease and schedule fit; once check-in data exist, item-based collaborative filtering adds habits that similar users kept (`recommenderlab`)	Nightly; after onboarding
P4	Habit stacking	Apriori rules "users who keep habit A also keep habit B" decide the next habit to propose (`arules`)	Weekly
P5	User personas	k-means clustering of at-home hours and check-in behaviour sets default reminder times for new users (`stats`, `cluster`)	Weekly
P6	Evaluation	Precision@5 (O1), back-test of P2 (O2), paired test of check-in rates before and after reminders stop (O3) (`recommenderlab`, `stats`)	Every sprint

IV. Architecture

The native iOS app (SwiftUI) keeps the profile, check-ins and the latest forecast in a local database (SwiftData), so it works offline and schedules the Green Hours reminders on the phone. It talks over REST to a FastAPI back-end, which stores the data shared between users (accounts, WG groups, model results) in PostgreSQL and calls the R service (Fig. 1). To keep the app's own footprint low, R runs as nightly or weekly batches instead of per request, and the back-end serves the stored results. The footprint is measured from Sprint 4 with the Xcode Energy Gauge, Docker statistics and the R packages `bench` and `profvis`, and measured again in every sprint.

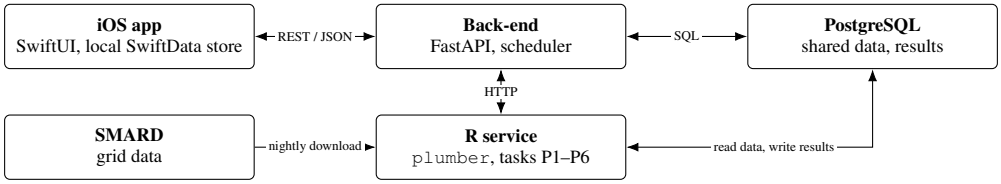


Figure 1. Architecture. All server components run in Docker containers; GitHub Actions runs the tests on every pull request.

V. Project Plan

The project runs as five Scrum sprints [11], tracked in Jira (project `HEC`). Sprint 5 is split into 5a and 5b so that each of the six members is Scrum Master once and Product Owner once (Table III).

TABLE III
SPRINTS AND SCRUM ROLES

#	Dates	Main results	Scrum Master	Product Owner
1	26 Oct – 6 Nov	Personas, user stories, Jira board, team contract, first SMARD data	M. Uddin	A. Newaz
2	9 – 20 Nov	Revised proposal, Figma wireframes, architecture, UML diagrams, student survey	M. Ambulkar	M. S. Zafar
3	23 Nov – 4 Dec	Back-end, database, R tasks P1–P2, first screens	M. R. H. Tanvir	N. Alam
4	7 – 18 Dec	R tasks P3–P5, all eleven screens, green baseline for O7	N. Alam	M. Ambulkar
5a	4 – 14 Jan	Two-week pilot with the team, optimisation	M. S. Zafar	M. Uddin
5b	15 – 25 Jan	Evaluation of O1–O7, final report and delivery	A. Newaz	M. R. H. Tanvir

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